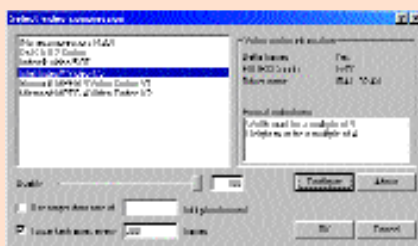


Lights, Camera, Action!

For capturing video we will use VirtualDub with a Web cam. VirtualDub is a very powerful freeware and allows better control over the capture process than most bundled software. Launch VirtualDub and click **File > Capture AVI** to switch to capture mode. You should set the resolution from **Video > Format** before you start capturing. Also, unless you are recording an interview, disable Capture audio from **Capture > Settings**. We will capture audio separately and merge it into the movie. Here you can also specify the capture frame rate. Align the camera to set the



Open Video > Compression to choose a codec to compress the captured video

frame you want to shoot and start capturing by clicking **Capture > Capture Video**. Hit [Esc] to stop recording.

Dropped frames: If your PC is slow, it may not process every frame sent by the capture device. To keep up with the incoming video, it simply drops the unprocessed frame and fills it with the preceding frame instead. This appears as jerks in the video. When capturing video, keep an eye on the number of frames dropped to see how smooth or jerky the video will be.

Sound settings: When recording sound separately from the video, you should use high quality settings—record 16-bit stereo sound at 44,100 Hz. Unlike with video, computers today have no problem processing high-quality sound. For the final video, though, scale it down to 8-bit stereo at 22,050 Hz. The sound will still be clear and the files will be much smaller.

Interlacing: Interlacing increases the visible resolution of a video, since two frames interleave to make one image. Interlaced capture with low-end equipment can result in a hazy picture, especially when recording movement. The camera sends two frames sequentially, so the two images that

make up one frame on the output may not match. Also, the computer needs to process more frames when capturing interlaced video. Besides, computer-generated graphics are non-interlaced and when merged with an interlaced video, may not render well. Hence, it is a better idea to always use a non-interlaced video.

Special effects: Video editing software such as VirtualDub (www.virtualdub.org) allows you to add some basic effects to the video. With a few combinations of filters, you can produce some really interesting effects. Apart from the bundled filters (see table, 'VirtualDub Filters'), you can also download a lot of filters from the Internet—www.virtualdub.org itself has a huge list you can choose from. It is best to apply effects filters after you've cleaned up

Let there be Light

Beware of Web cams. Most of them dynamically adjust brightness and contrast settings depending on the ambient light. This works well in a static frame, but if you are moving the camera, it can produce a few seconds of very dark or very bright video if the lighting is not even.

SoundForge 6.0, Virtual-Dub 1.4.10, Avi Assembler 1.2, DivX 5.02 Codec
Find them in Software > Multimedia on the Mindware CD. Plus, video clips in Surge > Samples on the Playback CD for you to experiment with



the video or at the last stage when you are generating the final video. The more filters you add, the slower will be the processing, though.

Video compression: Video compression codecs are not perfect—they cannot output exactly the same image as the original. The codec tries to estimate what differences would be invisible to the eye and removes them. The advantage is you can control how big the video file gets by changing its visible quality. Compression artefacts (noise in the video) result from high compression and the amount of noise varies across algorithms.

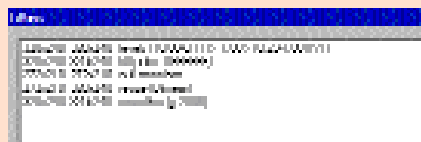
DivX is the best compression format available. It compresses video to ridiculously small sizes without too much loss in quality. DivX 5, the latest version, uses

Variable Bitrate (VBR), ensuring the smallest possible size with the highest possible quality for each frame. Several people have been developing the DivX codec for a while now and it is actually available in many different flavours. The codec can be downloaded from www.divx.com. (For more information on DivX, refer to Digit, March 2002.)

On the Editing Table

After the shoot, you will want to edit the video to remove all the unwanted parts and resample the image resolution and quality. VirtualDub displays two panes, one for the source video and the other for the same with filters applied. To cut unwanted frames out from the video, press the [Home] key where you want to start cutting the video and the [End] key where this ends. Hit [Delete] to cut out the marked area.

VirtualDub has a lot of filters to cleanup the video too (*see table, 'VirtualDub Filters'*). Select the portion of the video you want to clean up by clicking **Video > Select range**.



Filters in VirtualDub are applied in the order in which they are listed. Rearranging the order may yield totally different effects and also affects the processing time

You can add filters to process the video from **Video > Filters**. Click Add to add and configure the filters. Each filter will bring up its dialog box where you can configure it. Most of

them allow you to preview the effect of the filter on the fly. With each filter you add, the preview will include effects of all preceding filters. You can rearrange the order of the filters once you have added them.

You can also crop the video by clicking the Cropping button. VirtualDub does not allow you to crop a video without a filter, but you can use the Null Transform filter, which does not affect the video at all, but just provides an interface to allow cropping. Once you have set up all filters, click **Video > Full processing mode**—the filters will not be applied without this—and save the video.